

Department of Computer Science & Engineering

Software Engineering

SE Lab – 3: Architecture Selection & Component Modelling

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Remote Team Time-Tracking & Project Approver

Architecture Selection

I chose Layered Architecture for the Remote Team Time-Tracking & Project Approver system.

Architectural Choice

The system is divided into a Presentation Layer, Business Layer, and Integration/Data Layer. The Presentation Layer contains the Developer / Manager Dashboard. The Business Layer contains Time Tracking, Timesheet Approval Workflow, and Budget Burn Analytics. The Integration/Data Layer contains Jira Integration, the Timesheet & Project Database, and Authentication / RBAC, with Jira represented as an external system.

Reason 1 - Separation of Concerns

The assigned system performs distinct functions: developers log time against Jira tasks and submit timesheets, managers review and approve/reject timesheets, and project information is used to monitor budget burn. Separating presentation, business logic, and integration/data responsibilities keeps these functions organized and makes the system easier to maintain.

Reason 2 - Centralized Data and Workflow Management

The Timesheet & Project Database provides a common data layer for timesheet records, approval status, project information, and budget-related data. The Timesheet Approval Workflow and Budget Burn Analytics components can therefore use controlled business and data interfaces instead of directly accessing implementation details.

Security Advantage

The dedicated Authentication / RBAC Component can control access according to the user's role. This allows Remote Developers and Engineering Managers to be given different permissions for operations such as logging/submitting time, reviewing or approving timesheets, and viewing project budget information.

Performance Benefit

Separating Budget Burn Analytics into its own business component allows project burn information to be calculated and retrieved through focused queries rather than mixing the work with time-entry and approval operations. This reduces unnecessary processing in other components and supports responsive dashboard updates.

Conclusion

Layered Architecture provides a clear separation between the user interface, core business workflows, and Jira/database/security integrations. It matches the functional responsibilities represented in the use-case model while providing a structured architecture for the assigned system.